Forensic Analysis of the POLARIS Stage8.17A Failure: Atmospheric Mechanics, Disagreement Pitfalls, and Physical Pathways for Downwelling Shortwave Forecasting at Mount Nuolja

Forensic Deconstruction of the Stage8.17A Real-Data Failure

The introduction of multi-model scalar disagreement metrics in POLARIS Stage8.17A degraded downwelling surface shortwave irradiance (\text{SW_IN}) forecasts across the out-of-distribution (ODD) validation dataset. The candidate feature pipeline integrated two scalar fields calculated between the deterministic AROME-Arctic run and the MetCoOp Ensemble Prediction System (MEPS) control run:
\text{wind_disagreement_700} = \sqrt{(\text{AROME_U700} - \text{MEPS_U700})^2 + (\text{AROME_V700} - \text{MEPS_V700})^2} \text{LWP_disagreement} = \left\vert{} \text{AROME_LWP} - \text{MEPS_LWP} \right\vert{}
Out-of-distribution evaluation demonstrated an unambiguous drop in predictive skill. The control model established a baseline Mean Absolute Error (MAE) of 36.211027 \text{ W/m}^2, whereas the candidate model degraded to an MAE of 36.426418 \text{ W/m}^2, resulting in a net error increase of +0.215391 \text{ W/m}^2. A non-parametric bootstrap analysis established that this degradation was statistically robust rather than an artifact of sampling variability: the bootstrap mean degradation was +0.177449 \text{ W/m}^2, with a 95% bootstrap confidence interval spanning [-0.1127, +0.4749] \text{ W/m}^2 and an improvement probability of only 11.81\%.

Experiment Configuration	Surface \text{SW_IN} MAE (\text{W/m}^2)	\Delta \text{MAE} (\text{W/m}^2)	Bootstrap Mean (\text{W/m}^2)	Bootstrap 95% CI (\text{W/m}^2)	Improvement Probability (%)
Stage8 Control Baseline	36.211027	Reference	Reference	Reference	Reference
Stage8.17A Candidate Model	36.426418	+0.215391	+0.177449	[-0.1127, +0.4749]	11.81\%

Diagnostic forensics reveal that the candidate architecture failed due to four compounding statistical and atmospheric factors:
The foremost statistical defect was a near-zero correlation between the scalar disagreement features and the signed forecast residuals (y - \hat{y}). Because both metrics rely on unsigned Euclidean norms or absolute differences, they discard all information regarding the sign and vertical direction of the physical discrepancy. In atmospheric radiative transfer, knowing that two

models diverge in cloud liquid water path provides no actionable constraint on whether the control model has over- or under-predicted downwelling flux. Scalar disagreement acts as an uncalibrated proxy for dispersion rather than a directional error-correction operator.
The scalar disagreement features exhibited severe mutual collinearity with existing predictors in the Stage8 feature matrix. Predictors such as MEPS-derived liquid water path, total cloud cover, surface relative humidity, and isobaric wind speeds already populate the model's feature space. Adding difference magnitudes between two models sharing nearly identical physics cores introduced collinear noise without providing orthogonal variance.
Model degradation was not confined to isolated atmospheric regimes; instead, candidate error grew monotonically across every decile of the disagreement distribution. This behavior confirms that the machine learning system could not map scalar disagreement to a monotonic correction function. Instead, the features acted as non-informative perturbations that destabilized gradient splitting during tree induction.
The physical damage was heavily concentrated in high-solar, high-disagreement clear-sky events. During periods of high solar elevation, potential surface shortwave irradiance reaches peak seasonal values. In these cases, minor spatial displacement or slight timing offsets between AROME and MEPS created large absolute differences in local liquid water path and 700 \text{ hPa} wind speed. The machine learning model mapped large scalar disagreements to increased cloud attenuation risk, causing it to artificially hedge its predictions downward. Consequently, the candidate model predicted spurious cloud attenuation during verified clear-sky conditions, incurring substantial absolute error penalties at the top of the diurnal insolation curve.

Atmospheric Mechanics of Surface Shortwave Irradiance at Mount Nuolja

Topoclimatic Setting and Radiative Regimes

Mount Nuolja (\text{Njull\acute{a}}) rises to an elevation of 1,169 \text{ m} above sea level in the subarctic Scandes of northern Sweden (68.36^\circ \text{N}, 18.79^\cir[span_0](start_span)[span_0](end_span)[span_2](start_span)[span_2](end_span)[span_4](start_span)[span_4](end_span)c \text{E}). The meteorological station is situated near the tree line at approximately 900 \text{ m} above sea level, perched directly above the southern shore of Lake Torneträsk (341 \text{[span_6](start_span)[span_6](end_span)[span_8](start_span)[span_8](end_span)[span_10](start_span)[span_10](end_span)[span_12](start_span)[span_12](end_span) m} a.s.l.). The local radiation balance is dictated by complex terrain interactions coupled with high-latitude solar geometry.
Mount Nuolja is positioned in the immediate lee of the Scandes mountain range, which forms a major topographical barrier to prevailing maritime westerly flows from the North Atlantic. When moist, maritime air masses cross the Scandes, intense orographic lifting triggers precipitation on the windward Norwegian slopes. As the depleted air mass spills eastward across the crest and descends into the Torneträsk basin, forced adiabatic subsidence causes rapid warming and drying.
This localized föhn effect produces the well-documented "Blue Hole of Abisko," a persistent mesoscale opening in the regional cloud shield that gives the Abisko basin an exceptionally arid

microclimate, with annual precipitation near 300[span_57](start_span)[span_57](end_span)[span_58](start_span)[span_58](end_span)[span_59](start_span)[span_59](end_span)[span_60](start_span)[span_60](end_span)[span_61](start_span)[span_61](end_span)[span_62](start_span)[span_62](end_span) \text{ mm}. During westerly flow, local clearings frequently develop over Nuolja while surrounding regional NWP models predict continuous stratiform cloud cover.

A distinct, opposing topoclimatic regime emerges during periods of weak synoptic forcing or low-level easterly flow channeled along Lake Torneträsk. Radiative cooling and surface moisture pooling in the valley basin lead to the development of dense Arctic boundary-layer temperature inversions. These inversions frequently trap thick stratus and fog within the lowest hundreds of meters of the valley atmosphere.

Because the Nuolja observation platform sits at 900 \text{ m} above sea level, it often protrudes above this shallow valley inversion into cloud-free skies, while lower-elevation terrain remains occluded. Conversely, when the inversion layer deepens above 1,000 \text{ m}, the station becomes embedded in the dense cloud deck. The surface shortwave irradiance measured at Nuolja is therefore governed by sharp vertical boundary-layer transitions and localized orographic subsidence rather than broad regional cloud fields.

Solar Slant-Path Geometry Versus One-Dimensional Columnar Approximations

Downwelling surface shortwave irradiance (\text{SW_IN}) recorded by a horizontal pyranometer represents the sum of direct normal irradiance (\text{DNI}) projected onto the horizontal surface and diffuse horizontal irradiance (\text{DHI}):

\text{SW_IN} = \text{DNI} \cdot \cos\theta_z + \text{DHI}

where \theta_z is the solar zenith angle. The direct solar beam is attenuated according to the Beer-Lambert-Bouguer formulation:

\text{DNI} = I_0 \cdot \exp\left(-\frac{\tau_{\text{slant}}}{\cos\theta_z}\right)

where I_0 is extraterrestrial irradiance and \tau_{\text{slant}} is the integrated optical thickness along the line of sight from the ground instrument to the solar disc.

At subarctic latitudes (68.36^\circ \text{N}), the solar zenith angle is large throughout the year, exceeding 50^\circ at local noon during midsummer and remaining above 70^\circ to 85^\circ during spring and autumn. Under such oblique illumination, the atmospheric volume that intercepts the direct solar beam is displaced horizontally by several kilometers relative to the surface pyranometer. For a cloud layer situated at an altitude z_c above the ground station, the horizontal displacement vector (\Delta x, \Delta y) is given by:

\Delta x = z_c \cdot \tan\theta_z \cdot \sin\phi_s \Delta y = z_c \cdot \tan\theta_z \cdot \cos\phi_s

where \phi_s represents the solar azimuth angle relative to south. For a low-to-mid-level cloud layer 2,500 \text{ m} above the terrain at a solar zenith angle of \theta_z = 65^\circ, the horizontal displacement between the surface station and the intercepting cloud volume is:

\Delta r = 2.5 \cdot \tan(65^\circ) \approx 5.36 \text{ km}

In numerical weather prediction grids with a horizontal grid spacing of 2.5 \text{ km}—such as MEPS and AROME-Arctic—the cloud volume controlling direct beam attenuation is situated two to three grid cells away from the single vertical grid column directly above the sensor. Evaluating cloud presence, liquid water path (\text{LWP}), or optical thickness strictly within the zenith column above Mount Nuolja introduces an unavoidable geometric phase error.

A pyranometer on Nuolja can register clear-sky irradiance if an upstream gap exists along the

slant path, even if an overcast cloud layer sits directly overhead. Conversely, a small cloud patch located 5 \text{ km} up-sun will cast a shadow over the sensor while the zenith column remains clear. Standard one-dimensional columnar feature extractions fail to capture this three-dimensional geometric reality.

Hierarchy of Latent Cloud States

The physical variables governing surface shortwave radiation at Mount Nuolja can be arranged into an objective hierarchy based on their direct causal influence on downwelling solar flux.

Rank	Latent Atmospheric State	Governing Physical Mechanism	Impact on Surface \text{SW_IN}
1	**3D Slant-Path Position**	Intersection of hydrometeors with the solar line of sight.	Determines direct beam extinction versus clear-sky transmission.
2	**Slant Optical Thickness (\tau)**	Mie scattering and absorption integrated along the slant path.	Governs the exponential extinction rate of direct solar insolation.
3	**Cloud Thermodynamic Phase**	Specific extinction cross-sections: liquid (r_{\text{eff}} \approx 5\text{--}10\ \mu\text{m}) vs. ice (r_{\text{eff}} \ap[span_87](start_span)[span_87](end_span)[span_89](start_span)[span_89](end_span)prox 30\text{--}80\ \[span_137](start_span)[span_137](end_span)[span_138](start_span)[span_138](end_span)[span_139](start_span)[span_139](end_span)[span_140](start_span)[span_140](end_span)mu\text{m}).	Liquid clouds attenuate shortwave flux far more efficiently than ice per unit mass.
4	**Terrain-Induced Vertical Motion**	Föhn subsidence dissolving clouds vs. forced orographic lift.	Controls local cloud dissolution in the Abisko "Blue Hole".
5	**Boundary-Layer Inversion Height**	Capping inversion height relative to the 900 \text{ m} station elevation.	Determines whether the sensor is above valley fog or embedded in stratus.
6	**Condensate Advective Velocity**	Wind speed and direction within the	Dictates cloud transit speed and surface

Rank	Latent Atmospheric State	Governing Physical Mechanism	Impact on Surface \text{SW_IN}
		cloud hydrometeor layer.	shadow translation rates.
7	**Cloud Microphysical Lifetime**	Droplet collision-coalescence, autoconversion, and ice glaciation.	Regulates the temporal persistence and decay of cloud optical mass.
8	**Zenith 1D Columnar Cloud Presence**	Total hydrometeor mass directly overhead in a single vertical grid box.	Uncoupled from direct beam extinction under high-latitude slant geometry.

Structural Dissection of AROME-Arctic Versus MEPS

Shared Model Foundations and Domain Discrepancies

To understand why multi-model scalar disagreement degraded the POLARIS pipeline, the underlying numerical architectures of AROME-Arctic and MEPS must be examined. Both systems are developed within the international ACCORD consortium and share identical dynamical cores and atmospheric physics parameterizations.

Structural Feature	AROME-Arctic	MetCoOp Ensemble Prediction System (MEPS)
Consortium Baseline	ACCORD / HARMONIE-AROME	ACCORD / HARMONIE-AROME
Horizontal Resolution	2.5 \text{ km} grid spacing	2.5 \text{ km} grid spacing
Vertical Discretization	65 vertical levels (terrain-following hybrid pressure)	65 vertical levels (terrain-following hybrid pressure)
Microphysics Scheme	ICE3 1-moment bulk scheme with Arctic OCND2 options	ICE3 1-moment bulk scheme with Arctic OCND2 options
Radiation Scheme	IFS cycle 25R1 (Morcrette SW 6-band, RRTM LW 16-band)	IFS cycle 25R1 (Morcrette SW 6-band, RRTM LW 16-band)
Surface Core	SURFEX platform (ISBA diffusion soil and snow scheme)	SURFEX platform (ISBA diffusion soil and snow scheme)
Operational Domain	Northern Scandinavia, Svalbard, Barents Sea	Fennoscandia, Baltic Sea, southern Finland, Baltic States
Nuolja Domain Placement	Immediate southern boundary fringe	Central interior of the integration domain
Data Assimilation	3-hourly 3D-Var, Arctic satellite radiances, sea-ice fields	3-hourly 3D-Var, dense Nordic radar, SYNOP, Mode-S winds
Operational Architecture	High-resolution deterministic run	30-member convection-permitting ensemble

Because both systems operate at an identical 2.5 \text{ km} grid spacing and use matching ICE3/OCND2 microphysics and SURFEX surface schemes, AROME-Arctic possesses no superior horizontal resolution, more detailed digital elevation model, or more advanced cloud

microphysics scheme over Mount Nuolja compared to MEPS.
The structural divergence between the two models arises from domain geometry and data assimilation streams. Mount Nuolja is situated near the southern lateral boundary of the AROME-Arctic domain, where numerical solutions are influenced by lateral boundary condition blending with ECMWF IFS fields. Conversely, Nuolja is centered within the MEPS operational domain, well away from boundary buffer zones. Furthermore, MEPS assimilates a dense regional network of radar reflectivities, commercial Mode-S aircraft observations, and Fennoscandian surface SYNOP stations, whereas AROME-Arctic prioritizes high-latitude maritime soundings and sea-ice boundary conditions.

Physical Etiology of the Disagreement Metric Failure

Because AROME-Arctic and MEPS share identical model physics, any divergence between their outputs over Mount Nuolja does not reflect physical model uncertainty or structural diversity. Instead, their disagreement is dominated by spatial displacement jitter and lateral boundary phase lags.
In convection-permitting NWP models (2.5 \text{ km} resolution), fine-scale cloud structures are subject to the double penalty problem. A localized mountain cloud patch shifted by only 5 \text{ km} (two grid points) produces a large absolute error in grid-point \text{LWP}:
\left\vert{} \text{AROME_LWP} - \text{MEPS_LWP} \right\vert{} = \left\vert{} \text{LWP}_{\text{cloud}} - 0 \right\vert{} = \text{LWP}_{\text{cloud}}
even when both models accurately simulate the broader mesoscale cloud field over the Scandes. Because the scalar formulation \left\vert{} \text{AROME_LWP} - \text{MEPS_LWP} \right\vert{} removes the spatial gradient and sign, the machine learning model cannot determine which forecast is physically offset.
Furthermore, because lateral boundary condition updates arrive on different schedules and from different grid interpolations, frontal boundaries and short-wave troughs traverse the Scandinavian mountains with slight phase lags between the two systems. The resulting scalar differences represent transient temporal offsets rather than structural divergence in cloud morphology.
When the machine learning pipeline ingested these scalar disagreement metrics, it learned an uncalibrated hedging heuristic: whenever disagreement was high, it reduced predicted surface shortwave irradiance across all regimes. In high-insolation clear-sky cases, minor phase or spatial offsets generated large disagreement values, prompting the model to falsely predict cloud attenuation and producing the observed degradation in ODD performance.

Kinematic Transport, Layer Dynamics, and Terrain Interactions

Decoupling of 700 hPa Flow from Subarctic Cloud Decks

The candidate metric \text{wind_disagreement_700} quantified vector wind differences at the 700 \text{ hPa} isobaric surface. Under standard atmospheric conditions, 700 \text{ hPa} corresponds to approximately 3,000 \text{ m} a.s.l. Mount Nuolja's observing station sits at 900 \text{ m} a.s.l. (approximately 910\text{--}890 \text{ hPa}), its summit reaches 1,169 \text{ m} a.s.l. (~~880 \text{ hPa}), and Lake Torneträsk sits at 341 \text{ m} a.s.l. (~~970 \text{ hPa}).

The cloud decks that govern surface solar irradiance at Nuolja—including low-level mixed-phase stratocumulus, valley stratus, and orographic cap clouds—typically reside between the surface and 2,000 \[span_249](start_span)[span_249](end_span)[span_250](start_span)[span_250](end_span)[span_251](start_span)[span_251](end_span)[span_252](start_span)[span_252](end_span)text{ m} a.s.l. (950\text{--}800 \text{ hPa}). The 700 \text{ hPa} level is situated roughly 1,800 \text{ m} above the Nuolja station, placed well within the free troposphere.
In subarctic mountain terrain, strong static stability and surface-based temperature inversions frequently decouple boundary-layer valley air from the free troposphere above. Free-tropospheric westerly winds at 700 \text{ hPa} often blow directly across the mountain barrier while the lower-level air in the Torneträsk basin remains calm or drains westward as a localized density current. Evaluating wind disagreement at 700 \text{ hPa} therefore measures kinematic differences in an atmospheric layer that is physically detached from the low-level cloud hydrometeors attenuating surface solar radiation.

Condensate-Weighted Advection and the Advective Gradient Dilemma

Advecting cloud hydrometeors requires extracting the horizontal wind vector specifically within the vertical layers containing cloud water and ice. The vertically integrated, condensate-mass-weighted advection vector \vec{V}_{\text{steer}} = (U_{\text{steer}}, V_{\text{steer}}) is defined as:
U_{\text{steer}} = \frac{\int_{p_s}^{p_{\text{TOA}}} \left[q_l(p) + q_i(p) \right] U(p) \, dp}{\int_{p_s}^{p_{\text{TOA}}} \left[q_l(p) + q_i(p) \right] dp[span_7](start_span)[span_7](end_span)[span_9](start_span)[span_9](end_span)[span_11](start_span)[span_11](end_span)[span_13](start_span)[span_13](end_span)} V_{\text{steer}} = \frac{\int_{p_s}^{p_{\text{TOA}}} \left[q_l(p) + q_i(p) \right] V(p) \, dp}{\int_{p[span_23](start_span)[span_23](end_span)[span_27](start_span)[span_27](end_span)[span_31](start_span)[span_31](end_span)_s}^{p_{\text{TOA}}} \left[q_l(p) + q_i(p) \right] dp}
where q_l(p) and q_i(p) are specific liquid and ice water mass mixing ratios, and p_s is surface pressure.
While condensate-weighted wind fields provide a more accurate representation of cloud-layer transport than arbitrary isobaric surfaces, computing scalar disagreement between two models remains fundamentally flawed. If Model A predicts a boundary-layer stratocumulus deck at 900 \text{ hPa} and Model B predicts a cirrus deck at 400 \text{ hPa}, their respective condensate-weighted winds will sample entirely different vertical layers. The resulting vector difference \left\Vert{} \vec{V}_{\text{steer}, A} - \vec{V}_{\text{steer}, B} \right\Vert{} will reflect vertical cloud placement differences rather than kinematic steering divergence.
More fundamentally, local wind vectors cannot resolve cloud displacement without knowledge of the upstream spatial gradient of cloud water (\vec{V} \cdot \nabla_{\!H} \text{LWP}). In an advective transport equation:
\frac{\partial \, \text{LWP}}{\partial t} = - \vec{V} \cdot \nabla_{\!H} \text{LWP} + S_{\text{micro}}
where S_{\text{micro}} represents microphysical sources and sinks, the local rate of change is driven by the spatial gradient. If the upstream field is uniform (\nabla_{\!H} \text{LWP} = 0), cloud water over the station remains unchanged regardless of wind speed. Conversely, strong winds blowing across a sharp cloud boundary produce rapid local cloud transitions. Because the candidate Stage8.17A features omitted horizontal spatial gradients, the machine learning system was deprived of the physical information required to infer cloud displacement.

Froude Number Dynamics and the Abisko Rain Shadow

Cloud formation and dissolution at Mount Nuolja are governed by the interaction between low-level atmospheric flow and the complex topography of the Scandes. For stratified flow encountering a topographic barrier of height H_m \appr[span_129](start_span)[span_129](end_span)ox 1,200 \text{ m}, the kinematic regime is governed by the mountain Froude number (Fr):
Fr = \frac{U_\perp}{N \cdot H_m}
where U_\perp is the cross-barrier wind velocity perpendicular to the Scandes axis (westerly flow), and N is the Brunt-Väisälä buoyancy frequency:
N = \sqrt{\frac{g}{\theta_v} \frac{\partial \theta_v}[span_263](start_span)[span_263](end_span){\partial z}}
representing atmospheric static stability.
When Fr \ge 1, cross-barrier flow possesses sufficient kinetic energy to surmount the ridge line. Moist air ascends the western windward slopes in Norway, precipitates out, and undergoes forced adiabatic subsidence as it spills into the Torneträsk basin. Downward vertical motion warms the descending air at the dry adiabatic lapse rate (\Gamma_d \approx 9.8 \text{ K/km}), lowering relative humidity and dissolving cloud condensate. This regime establishes the "Blue Hole," producing clear skies and elevated shortwave insolation at Mount Nuolja.
When Fr \ll 1, the low-level flow lacks kinetic energy to cross the terrain barrier, leading to upstream blocking and stagnant conditions in the Torneträsk valley. In this regime, localized radiative cooling dominates, trapping dense fog and stratus beneath the valley inversion.
The scalar metric \text{wind_disagreement_700} obscures this physical behavior. By stripping away wind direction, cross-barrier sign, and static stability, the metric cannot differentiate between a high-irradiance föhn clearing event and a low-irradiance valley trapping regime.

Multi-Hour Temporal Trajectories and Point-Observation Ambiguity

Tracking disagreement trajectories across a 3- to 6-hour window (\Delta \text{LWP}_{t}, \Delta \text{LWP}_{t-1}, \dots, \Delta \text{LWP}_{t-6}) can help identify temporal phase shifts in synoptic frontal passages. For example, a time-lagged cross-correlation:
R_k = \sum_{m} \text{LWP}_{\text{AROME}}(t[span_247](start_span)[span_247](end_span)_m) \cdot \text{MEPS_LWP}(t_m + k)
can detect whether one model consistently leads or lags the other during frontal transitions. However, tracking a temporal trajectory at a single spatial grid point cannot resolve the underlying physical mechanism. At a single station, an observed decline in cloud liquid water over three hours can result from spatial advection of a clear-air pocket, in-situ föhn evaporation driven by cresting westerly flow, or microphysical glaciation of supercooled liquid droplets into falling ice crystals. Without spatial field tracking across the surrounding terrain, point-based temporal trajectories remain physically ambiguous and cannot isolate advective displacement errors.

Evaluation and Ranking of Breakthrough Mechanisms

To identify research pathways that avoid the failure modes of Stage8.17A, five candidate mechanisms are evaluated across eight criteria:

- *Physical Plausibility:* Direct coupling to the governing radiative transfer equations and

mountain atmospheric dynamics.

- *Independence from Existing Stage8 Features:* Absence of mutual information overlap or collinearity with features already present in the POLARIS baseline.
- *Causal Availability:* Operational accessibility strictly prior to forecast issue time T_0, with zero data leakage.
- *Spatial Representativeness:* Ability to resolve the sub-grid topoclimatic scale of Mount Nuolja and the Torneträsk basin.
- *Temporal Resolution:* Frequency of updates matched to the operational forecast cycle and cloud field lifetimes.
- *Expected Information Novelty:* Delivery of orthogonal physical states not captured by single-point vertical extractions.
- *Implementation Feasibility:* Software and data engineering complexity within an operational inference pipeline.
- *Falsifiability:* Clear, quantifiable empirical conditions under which the hypothesis must be rejected.

Breakthrough Mechanism	Physical Plausibility	Feature Independence	Causal Availability	Spatial Representativeness	Temporal Resolution	Expected Info Novelty	Implementation Feasibility	Falsifiability	Final Ranking
1D Multi-Model Disagreement *(Stage8.17A Baseline)*	Low	Low	High	Poor	Low	None	High	High	**Rejected**
Condensate-Weighted Advective Disagreement	Moderate	Moderate	High	Moderate	Low	Moderate	Moderate	High	**Rank 4**
Direct Satellite Cloud Nowcasting (NWC SAF PPS)	High	Very High	Moderate	High	Irregular	Extreme	Low	High	**Rank 3**
Orographic Froude Regime	High	High	High	High	Moderate	High	High	High	**Rank 2**

Breakthrough Mechanism	Physical Plausibility	Feature Independence	Causal Availability	Spatial Representativeness	Temporal Resolution	Expected Info Novelty	Implementation Feasibility	Falsifiability	Final Ranking
Indexing									
3D Slant-Path Hydrometeor Ray-Tracing	Very High	High	High	Very High	Moderate	Very High	Moderate	High	**Rank 1**

Three Candidate Research Hypotheses

Hypothesis 1: Three-Dimensional Solar Slant-Path Hydrometeor Ray-Tracing

The physical mechanism governing direct surface shortwave irradiance is optical extinction along the slant path connecting the ground pyranometer to the solar disc, rather than hydrometeor accumulation within the vertical zenith column. At subarctic latitudes (68.36^\circ \text{N}), the solar ray intersects low-to-mid-tropospheric cloud layers between 3 \text{ km} and 12 \text{ km} horizontally displaced from the station along the solar azimuth. Tracing this line-of-sight vector through the 3D prognostic hydrometeor fields of the operational NWP model resolves upstream cloud interception and clear-sky gaps, directly mitigating the false-alarm hedging that degraded Stage8.17A.

The exact missing information is the horizontal spatial offset coordinate (\Delta x, \Delta y) and the integrated slant-path optical thickness \tau_{\text{slant}}:

\tau_{\text{slant}} = \int_0^{s_{\text{TOA}}} \left[\beta_{\text{ext}, l}(\vec{r}(s)) + \beta_{\text{ext}, i}(\vec{r}(s)) \right] ds \vec{r}(s) = \vec{r}_{\text{station}} + s \cdot \begin{bmatrix} \sin\theta_z \sin\phi_s \\ \sin\theta_z \cos\phi_s \\ \cos\theta_z \end{bmatrix}

where s is the path distance, \theta_z is the solar zenith angle, \phi_s is the solar azimuth, and \beta_{\text{ext}} is the volume extinction coefficient parameterized via prognostic cloud water mixing ratios (q_l, q_[span_293](start_span)[span_293](end_span)[span_294](start_span)[span_294](end_span)[span_295](start_span)[span_295](end_span)[span_296](start_span)[span_296](end_span)[span_297](start_span)[span_297](end_span)i) and effective droplet/crystal radii (r_{\text{eff}}).

The candidate data source consists of operational MEPS 3D vertical model-level prognostic hydrometeor fields (q_l, q_i, T, p) on hybrid levels across a local 50 \ti[span_298](start_span)[span_298](end_span)[span_299](start_span)[span_299](end_span)mes 50 \text{ km} bounding box centered over Mount Nuolja.

Causal feasibility is preserved. The 3D MEPS forecast fields are generated prior to inference time, and solar coordinates (\theta_z, \phi_s) are deterministic astronomical functions of station coordinates and timestamp.

Current POLARIS features cannot represent this state because they extract 1D columnar scalars from the single horizontal grid box containing the station, leaving the model blind to lateral cloud shadows or upstream slant-path clearings.

Supporting evidence is found across three-dimensional atmospheric radiative transfer literature, which demonstrates that 1D approximations incur massive instantaneous irradiance errors at large solar zenith angles. In satellite meteorology and solar power forecasting, parallax and slant-path geometric corrections are established operational requirements.
Evidence against the hypothesis centers on NWP spatial placement accuracy: if MEPS misplaces a synoptic cloud boundary by more than 15 \text{ km}, ray-tracing will sample the wrong atmospheric coordinates. Additionally, spatial 3D interpolation increases data ingestion overhead compared to single-point extractions.
The hypothesis would be falsified if the integration of \tau_{\text{slant}} fails to produce a statistically significant reduction in \text{SW_IN} MAE under broken cloud conditions with intermediate-to-high solar zenith angles (50^\circ \le \theta_z \le 80^\circ), or if its signed error residuals remain collinear with the 1D columnar baseline.
An experiment is justified based on clear physical causality and direct alignment with the high-solar clear-sky failure mode observed in Stage8.17A.

Hypothesis 2: Upstream Orographic Stratification and Froude Number Indexing

The physical mechanism controlling cloud presence at Mount Nuolja is the orographic transition between leeward adiabatic subsidence (the "Blue Hole" föhn clearing) and low-level boundary-layer trapping. Computing the upstream terrain-perpendicular mountain Froude number (Fr = U_\perp / N H_m) provides the machine learning system with an explicit index of the orographic flow regime. When Fr \ge 1 and westerly cross-barrier winds crest the Scandes divide, forced leeward subsidence warms and dries the sub-crest layer, dissolving clouds over the Abisko basin. When Fr \ll 1, flow blocking and stable thermal stratification trap low-level stratus and fog within the Torneträsk basin.
The exact missing information is the non-dimensional hydrodynamic flow regime parameter (Fr) and the lower-tropospheric inversion strength (\Delta \theta_v = \theta_{v, 85[span_270](start_span)[span_270](end_span)[span_272](start_span)[span_272](end_span)[span_274](start_span)[span_274](end_span)0[span_319](start_span)[span_319](end_span)[span_320](start_span)[span_320](end_span)} - \theta_{v, \text{sfc}}) upstream of the Scandes crest.
The candidate data source consists of upstream vertical profiles of horizontal wind velocity, temperature, and geopotential height extracted from MEPS along the Norwegian coastal mountain divide (e.g., 68.4^\circ \text{N}, 17.5^\circ \text{E}) across the 1000\text{--}700 \text{ hPa} pressure layers.
Causal feasibility is preserved, as upstream profiles are causally accessible from operational MEPS run cycles without latency penalties.
Current POLARIS features cannot represent this state because they utilize localized surface wind and temperature observations on Mount Nuolja. Local point winds are subject to severe sub-grid terrain sheltering, mechanical turbulence, and local slope circulations that decouple from the synoptic-scale orographic forcing across the Scandinavian barrier.
Supporting evidence is established by empirical topoclimatic studies of the Abisko rain shadow, which confirm that clear-sky frequency is strongly correlated with westerly cross-barrier wind speed and negatively correlated with low-level static stability. In mountain meteorology, the Froude number is the classical metric governing orographic cloud transitions.
Evidence against the hypothesis is that MEPS’s non-hydrostatic dynamical core already

computes the primitive equations across the Scandes at 2.5 \text{ km} resolution, meaning large-scale subsidence is partially reflected in its raw diagnostic output. If the machine learning model already extracts the resulting adiabatic drying, adding an explicit bulk Froude index may yield only modest incremental variance.
The hypothesis would be falsified if the Froude index exhibits near-zero mutual information with signed \text{SW_IN} forecast errors, or if it fails to improve MAE during verified high-wind westerly föhn events (U_{700} > 15 \text{ m/s}).
An experiment is justified as a secondary, low-complexity feature pipeline requiring only 1D upstream profile extractions.

Hypothesis 3: Polar-Orbiting Direct Satellite Nowcasting via NWC SAF PPS

The physical mechanism governing short-range cloud evolution involves complex microphysical phase transitions and boundary-layer turbulence that convection-permitting NWP models parameterize with notable uncertainty. Direct multispectral observations from polar-orbiting meteorological satellites (MetOp-B/C, NOAA-20/21) processed through the EUMETSAT NWC SAF Polar Platform System (PPS) deliver empirical, model-independent cloud properties—including Cloud Mask (CMA), Cloud Top Height (CTH), and Cloud Optical Thickness (COT)—across northern Scandinavia. Extrapolating the observed upstream cloud mask along layer-specific steering winds provides an empirical cloud boundary arrival constraint that bypasses the shared microphysical biases of AROME and MEPS.
The exact missing information is the true, observationally verified spatial position, cloud-top altitude, and optical thickness of cloud patches upwind of Mount Nuolja.
The candidate data source is the EUMETSAT NWC SAF PPS cloud product suite derived from AVHRR and VIIRS instruments, disseminated via the EUMETSAT Advanced Retransmission Service (EARS).
Causal feasibility is constrained but operational. At 68.36^\circ \text{N}, polar satellite overpasses occur every 50 to 100 minutes. Via EARS direct-readout stations in Tromsø and Kangerlussuaq, data latency is 15 to 30 minutes, allowing causal ingestion for nowcasting horizons spanning +1 \text{ to } +6 \text{ hours}.
Current POLARIS features cannot represent this state because they rely exclusively on NWP model predictions and point surface observations, incorporating no spatial remote-sensing observations of upstream cloud systems.
Supporting evidence indicates that satellite-based nowcasting schemes consistently outperform convection-permitting NWP models in short-term cloud cover prediction (0\text{--}3 \text{ hours}) because they are unaffected by model spin-up errors or microphysical parameterization drift.
Evidence against the hypothesis includes satellite retrieval errors over snow-covered mountain terrain, where passive optical and thermal sensors struggle to separate cold low clouds from frozen ground. Furthermore, polar overpasses are irregularly spaced in time, and clouds crossing the Scandes divide can rapidly evaporate due to leeward subsidence, degrading the skill of simple advective extrapolation. Data pipeline latency and operational complexity are also substantially higher than NWP-based post-processing.
The hypothesis would be falsified if satellite-advected cloud products fail to achieve a higher Critical Success Index (CSI) or lower \text{SW_IN} MAE than the raw MEPS forecast at a 1-hour lead time over Mount Nuolja.
An experiment is not justified for immediate implementation due to ingestion latency,

snow-masking vulnerabilities, and infrastructure complexity. It remains an exploratory Phase 2 objective.

Definitive Strategic Selection and Operational Directives

Based on rigorous assessment of physical causality, independence from existing Stage8 predictors, operational availability, spatial representativeness, and direct remediation of the failure modes documented in Stage8.17A, the single strongest research route is **Hypothesis 1: Three-Dimensional Solar Slant-Path Hydrometeor Ray-Tracing**.

The forensic evidence of Stage8.17A demonstrated that the candidate model failed because scalar disagreement acted as an unsigned, collinear noise parameter that forced the machine learning system to hedge downward during high-solar clear-sky regimes. High-latitude mountain environments like Mount Nuolja exhibit large solar zenith angles throughout the year, causing the direct solar beam to traverse cloud fields located several kilometers away from the station's vertical column.

By tracing the line of sight through the three-dimensional hydrometeor field of the operational MEPS model, the system replaces 1D vertical approximations with the true geometry of direct solar beam extinction. When the solar ray passes through an upstream gap in a broken cloud deck, the slant-path optical thickness accurately evaluates to zero, allowing the ML model to output full clear-sky irradiance and preventing the false-alarm attenuation that degraded the Stage8.17A candidate.

Hypothesis 1 utilizes existing MEPS 3D prognostic volumes, ensuring full operational causality without introducing the irregular latency, complex data infrastructure, and snow-cover retrieval errors associated with satellite nowcasting (Hypothesis 3). Furthermore, whereas orographic Froude indexing (Hypothesis 2) provides only a broad, macroscale regime indicator, Hypothesis 1 directly solves the deterministic geometric Beer-Lambert equation governing surface solar insolation.

No model experiment, parameter optimization, or production code modification is authorized by this analysis. Hypothesis 1 represents the sole justified scientific pathway for future development, and experimental implementation remains strictly deferred until authorized by the systems governance framework.

Works cited

1. Abisko Northern Lights Live – Radar and Forecast - OndaTravel, https://ondatravel.pl/en/northern-lights-radar-locations-and-forecasts/abisko-northern-lights-live-radar-forecast/ 2. Northern Lights in Abisko, Sweden | World's Best Aurora Destination, https://polarforecast.com/northern-light-forecast-abisko 3. THE BALTEX BRIDGE CAMPAIGN - Radiometrics, https://radiometrics.com/wp-content/uploads/2021/10/Crewell_BAMS_04.pdf 4. Abisko Stargazing Guide: Sweden's Clearest Arctic Sky, https://www.darkskiesatlas.com/destinations/abisko-stargazing 5. I put Abisko's cloud-busting weapon' to the test during a Sweden, https://www.space.com/northern-lights-abisko-sweden-blue-hole 6. Abisko - Grokipedia, https://grokipedia.com/page/Abisko 7. SOLAR RADIATION PRODUCTS FOR THE END-USERS, https://www.solarpaces.org/wp-content/uploads/SunUpReport_210419-V2-web.pdf 8. 2021 -

IEA-PVPS,
https://iea-pvps.org/wp-content/uploads/2021/06/IEA_PVPS_T16_Solar_Res_Handbook_20211028.pdf 9. Best Practices Handbook for the Collection and Use of Solar,
https://docs.nlr.gov/docs/fy21osti/77635.pdf 10. Observation Operator for Visible and Near-Infrared Satellite,
https://journals.ametsoc.org/view/journals/atot/31/6/jtech-d-13-00116_1.xml 11. amt-2024-40-AC1-supplement.pdf,
https://amt.copernicus.org/preprints/amt-2024-40/amt-2024-40-AC1-supplement.pdf 12. Report on the state of science and knowledge on atmospheric icing,
https://www.easa.europa.eu/en/downloads/143972/en 13. A Kilometer-Scale Coupled Atmosphere–Wave Forecasting System,
https://journals.ametsoc.org/view/journals/wefo/36/6/WAF-D-21-0065.1.pdf 14. An Integrated Approach for a Better Understanding of Clouds,
https://www.researchgate.net/publication/216681343_The_BALTEX_Bridge_Campaign_An_integrated_approach_for_a_better_understanding_of_clouds 15. The Cycle 46 Configuration of the HARMONIE-AROME Forecast, https://www.mdpi.com/2674-0494/3/4/18 16. The domain of the AROME-Arctic model. The shading over land,
https://www.researchgate.net/figure/The-domain-of-the-AROME-Arctic-model-The-shading-over-land-indicates-model-topography_fig1_354592641 17. Multi-domain dynamical graph training of Bris for high-resolution on, https://cesoc.net/wp-content/uploads/2025/07/all-abstracts.pdf 18. HIRLAM data assimilation: current status and vision,
https://events.ecmwf.int/event/428/contributions/4865/attachments/2977/5169/DA-WS_Randriamampianina.pdf 19. The HARMONIE–AROME Model Configuration in the ALADIN,
https://repositorio.aemet.es/bitstream/20.500.11765/7320/1/MWR-D-16-0417.pdf 20. MPACE poster - ACCORD, https://www.accord-nwp.org/IMG/pdf/mpace_poster_engdahl.pdf 21. NORA3: A Nonhydrostatic High-Resolution Hindcast of the North,
https://journals.ametsoc.org/view/journals/apme/60/10/JAMC-D-21-0029.1.xml 22. Impact of Atmospheric Motion Vectors (AMV) on rapid update cycling,
https://www.met.no/publikasjoner/met-report/met-report-2017/_/attachment/download/f73796bf-5916-4eca-841f-5d8315455c13:80e930cb50b76b794672c2949c0b37c624f4d6d9/met-report-04-2017.pdf 23. Guidelines for the preparation of final papers for EUMETSAT,
https://cgms-info.org/html/iww14/papers/IWW14_Session04_09_Randriamampianina_final.pdf 24. Observing System Experiments with an Arctic Mesoscale Numerical,
https://www.mdpi.com/2072-4292/11/8/981 25. special project progress report - ECMWF,
https://www.ecmwf.int/sites/default/files/special_projects/2023/spsehlam-2023-report3.pdf?49629= 26. Verification of Operational Weather Prediction Models March to May,
https://www.met.no/publikasjoner/met-info/_/attachment/inline/6365c354-95f2-4834-b325-ca0941fc72c1:21bfb6c2589ad24875b3baafe0a5096d4081e5e6/met-info-27-2026.pdf 27. Forecaster perspectives on model quality and verification at ... - ASR,
https://asr.copernicus.org/articles/23/9/2026/ 28. The AROME-France convective-scale operational model,
https://www.researchgate.net/publication/258537490_The_AROME-France_convective-scale_operational_model 29. Data from Abisko Scientific Research Station - Swedish polar,
https://www.polar.se/en/research-support/open-data/data-from-abisko-scientific-research-station/ 30. Use of CAMS near Real-Time Aerosols in the HARMONIE-AROME,
https://www.mdpi.com/2674-0494/3/2/8 31. NWC/PPS-High Resolution Winds "Basic AMV" output example in,
https://www.researchgate.net/figure/NWC-PPS-High-Resolution-Winds-Basic-AMV-output-exam

ple-in-the-region-EURON1_fig2_357528250 32. Aurora Sky Station - Tickets, Chairlift & Northern Lights Guide, https://www.kirunatours.com/travel-guide/aurora-sky-station-guide/ 33. glacial-lake outburst floods: Topics by Science.gov, https://www.science.gov/topicpages/g/glacial-lake+outburst+floods 34. Abisko - Swedish Lapland, https://www.swedishlapland.com/stories/skiing-in-abisko/ 35. (PDF) Satellite-based solar radiation mapping over complex terrain, https://www.researchgate.net/publication/258772460_Satellite-based_solar_radiation_mapping_over_complex_terrain_Validation_in_the_Alps_and_possible_improvements 36. Snow Cover Changes & Dynamics in Northern Sweden, https://www.diva-portal.org/smash/get/diva2:2067241/FULLTEXT01.pdf 37. Our latest publications - Department of Geosciences - UiO, https://www.mn.uio.no/geo/english/research/publications/ 38. Wind Product Demonstration - NWCSAF-PPS - SMHI, https://nwcsaf.smhi.se/pps_wind_product_demonstration.php 39. METreport No. 12/2018 - Meteorologisk institutt, https://www.met.no/publikasjoner/met-report/met-report-2018/_/attachment/download/5e28741e-ccb3-4dd9-8e12-fb131661fc13:fb74c03d869d5180e4de8ff90cda39cc03549e89/MET-report-12-2018.pdf 40. CLARA-A3: The third edition of the AVHRR-based CM SAF climate, https://essd.copernicus.org/articles/15/4901/2023/ 41. (PDF) Probabilistic Cloud Masking for the Generation of CM SAF, https://www.researchgate.net/publication/339419639_Probabilistic_Cloud_Masking_for_the_Generation_of_CM_SAF_Cloud_Climate_Data_Records_from_AVHRR_and_SEVIRI_Sensors 42. clara-a3 - CM SAF Web User Interface // DOI // Details, https://wui.cmsaf.eu/safira/action/viewDoiDetails?acronym=CLARA_AVHRR_V003 43. Benefits of VIIRS data for EUMETSAT user community - NESDIS, https://www.nesdis.noaa.gov/s3/2022-07/Sreerekha_Thonipparambil-VIIRS_Benefits_EUMETSAT.pdf 44. Scientific service framework for Copernicus sea and sea-ice surface, https://www-cdn.eumetsat.int/files/2025-04/NOV-FE-1333-NT-015_Technical-note-on-validation-data_v2.2.pdf 45. CLARA-A2: The second edition of the CM SAF cloud and, https://acp.copernicus.org/preprints/acp-2016-935/acp-2016-935-manuscript-version5.pdf